

House Price Prediction - Summary

Objective:

To predict house prices using machine learning techniques and identify the most influential factors affecting property prices.

Dataset:

Housing dataset containing 545 records and 13 features.

Models Used:

1. Linear Regression
2. Random Forest Regressor

Results:

Linear Regression:

MAE = 970043.40

RMSE = 1324506.96

R² Score = 0.6593

Random Forest:

MAE = 1021546.04

RMSE = 1400565.97

R² Score = 0.612

Key Findings:

- Area strongly influences house price.
- Bathrooms and stories increase property value.
- Air conditioning and preferred area positively impact prices.
- Furnishing status also contributes to valuation.

Conclusion:

Linear Regression achieved the best performance with an R^2 Score of approximately 66%. Property size and amenities were identified as the most significant factors affecting house prices.